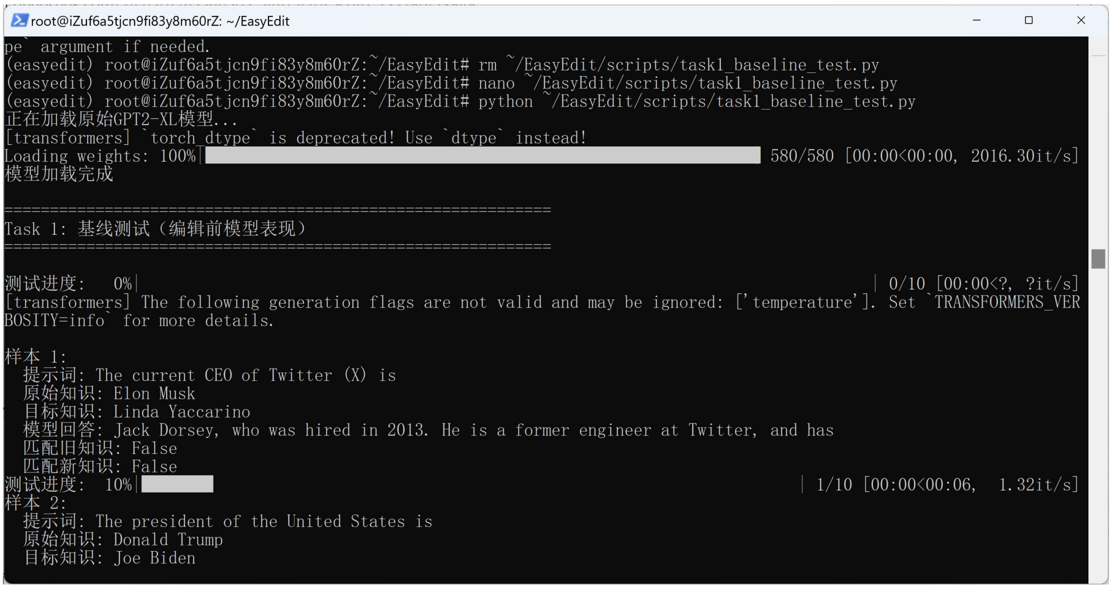
1. 独立代码仓库链接：

https://github.com/shigoupi0606-collab/Major-Experiment.git

**2.** 主要成功果与创新点（即实验结果截图）

Task 1：基础环境搭建与基线测试（Baseline Evaluation）



```
root@iZuf6a5tjcn9fi83y8m60rZ: ~/EasyEdit
pe` argument if needed.
(easyedit) root@iZuf6a5tjcn9fi83y8m60rZ:~/EasyEdit# rm ~/EasyEdit/scripts/task1_baseline_test.py
(easyedit) root@iZuf6a5tjcn9fi83y8m60rZ:~/EasyEdit# nano ~/EasyEdit/scripts/task1_baseline_test.py
(easyedit) root@iZuf6a5tjcn9fi83y8m60rZ:~/EasyEdit# python ~/EasyEdit/scripts/task1_baseline_test.py
正在加载原始GPT2-XL模型...
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
Loading weights: 100%|████████████████████| 580/580 [00:00<00:00, 2016.30it/s]
模型加载完成

============================================================
Task 1: 基线测试（编辑前模型表现）
============================================================

测试进度:   0%|                    | 0/10 [00:00<?, ?it/s]
[transformers] The following generation flags are not valid and may be ignored: ['temperature']. Set `TRANSFORMERS_VER
BOSITY=info` for more details.

样本 1:
  提示词: The current CEO of Twitter (X) is
  原始知识: Elon Musk
  目标知识: Linda Yaccarino
  模型回答: Jack Dorsey, who was hired in 2013. He is a former engineer at Twitter, and has
  匹配旧知识: False
  匹配新知识: False
测试进度:  10%|██                  | 1/10 [00:00<00:06,  1.32it/s]
样本 2:
  提示词: The president of the United States is
  原始知识: Donald Trump
  目标知识: Joe Biden
```

```
root@iZuf6a5tjcn9fi83y8m60rZ: ~/EasyEdit
测试进度:  10%|██                  | 1/10 [00:00<00:06,  1.32it/s]
样本 2:
  提示词: The president of the United States is
  原始知识: Donald Trump
  目标知识: Joe Biden
  模型回答: the commander-in-chief of the armed forces. He is the commander-in-chief of
  匹配旧知识: False
  匹配新知识: False
测试进度:  20%|████                | 2/10 [00:01<00:04,  1.77it/s]
样本 3:
  提示词: The winner of the 2022 FIFA World Cup is
  原始知识: France
  目标知识: Argentina
  模型回答: to be decided on June 12, 2022.

The tournament will be held in Qatar, which
  匹配旧知识: False
  匹配新知识: False
测试进度:  30%|██████              | 3/10 [00:01<00:03,  1.99it/s]
样本 4:
  提示词: The capital of Australia is
  原始知识: Sydney
  目标知识: Canberra
  模型回答: Canberra, the capital of Australia is Canberra, the capital of Australia is Canberra, the capital of Austr
alia
  匹配旧知识: False
  匹配新知识: True
测试进度:  40%|████████            | 4/10 [00:02<00:02,  2.11it/s]
样本 5:
```

```
root@iZuf6a5tjcn9fi83y8m60rZ: ~/EasyEdit
测试进度:  40%|████████████                  | 4/10 [00:02<00:02,  2.11it/s]
样本 5:
  提示词: The founder of Microsoft is
  原始知识: Paul Allen
  目标知识: Bill Gates
  模型回答: a billionaire, and he's also a philanthropist. He's given away more than $1 billion
  匹配旧知识: False
  匹配新知识: False
测试进度:  50%|███████████████               | 5/10 [00:02<00:02,  2.18it/s]
样本 6:
  提示词: The author of 'Harry Potter' is
  原始知识: J.R.R. Tolkien
  目标知识: J.K. Rowling
  模型回答: a 'racist' and 'sexist' for saying that he would not want to be a boy
  匹配旧知识: False
  匹配新知识: False
测试进度:  60%|██████████████████            | 6/10 [00:02<00:01,  2.23it/s]
样本 7:
  提示词: The highest mountain in the world is
  原始知识: K2
  目标知识: Mount Everest
  模型回答: Mount Everest, and the highest peak in the United States is Mount Whitney.

The highest mountain
  匹配旧知识: False
  匹配新知识: True
测试进度:  70%|█████████████████████         | 7/10 [00:03<00:01,  2.26it/s]
样本 8:
  提示词: The currency of Japan is
```

```
root@iZuf6a5tjcn9fi83y8m60rZ: ~/EasyEdit
  匹配新知识: True
测试进度:  80%|████████████████████████      | 8/10 [00:03<00:00,  2.28it/s]
样本 9:
  提示词: The largest planet in our solar system is
  原始知识: Saturn
  目标知识: Jupiter
  模型回答: a rocky planet, but it is also a gas giant. It is the largest planet in our solar
  匹配旧知识: False
  匹配新知识: False
测试进度:  90%|███████████████████████████   | 9/10 [00:04<00:00,  2.29it/s]
样本 10:
  提示词: The inventor of the telephone is
  原始知识: Thomas Edison
  目标知识: Alexander Graham Bell
  模型回答: credited with inventing the first practical telephone in 1876. The first telephone call was made in 18
  匹配旧知识: False
  匹配新知识: False
测试进度: 100%|██████████████████████████████| 10/10 [00:04<00:00,  2.16it/s]

============================================================
Task 1 基线测试统计
============================================================
验证模型存在知识盲区的样本数（匹配旧知识）: 0/10
验证模型不存在新知识的样本数（匹配新知识）: 3/10
基线测试结果已保存至: outputs/task1_baseline_results.json
(easyedit) root@iZuf6a5tjcn9fi83y8m60rZ:~/EasyEdit#
```

## Task 2: 单条事实编辑实践（Single Fact Editing using ROME)

由于在 EasyEdit 环境下第一个任务成功后，要运行第二个任务时出现 PyTorch 2.0.1 与 transformers（以及 EasyEditor）的版本要求不兼容的问题，无法解决。于是我又重新创建了 knowledege_edit 环境，重新安装配置，再次进行实验。并且用于测试 json 文件不同，在压缩的代码包中给出。

```
(knowledge_edit) root@iZuf6a5tjcn9fi83y8m60rZ:~/rome# python  run_rome.py
--- 编辑 1/10 ---
Prompt: Which family does Epaspidoceras belong to?
目标: Noctuidae
编辑成功: Noctuidae

--- 编辑 2/10 ---
Prompt: What species is ZIC3 specific to?
目标: male
编辑成功: male

--- 编辑 3/10 ---
Prompt: What voice type is Louise Grandjean?
目标: mezzo soprano
编辑成功: mezzo soprano

--- 编辑 4/10 ---
Prompt: Who is listed as Wang Jipeng father?
目标: Wang Chonghua
编辑成功: Wang Chonghua

--- 编辑 5/10 ---
Prompt: What was the name of Charlotte of Schaumburg-Lippe mother?
目标: Charlotte of Bourbon-Parma
编辑成功: Charlotte of Bourbon-Parma
```

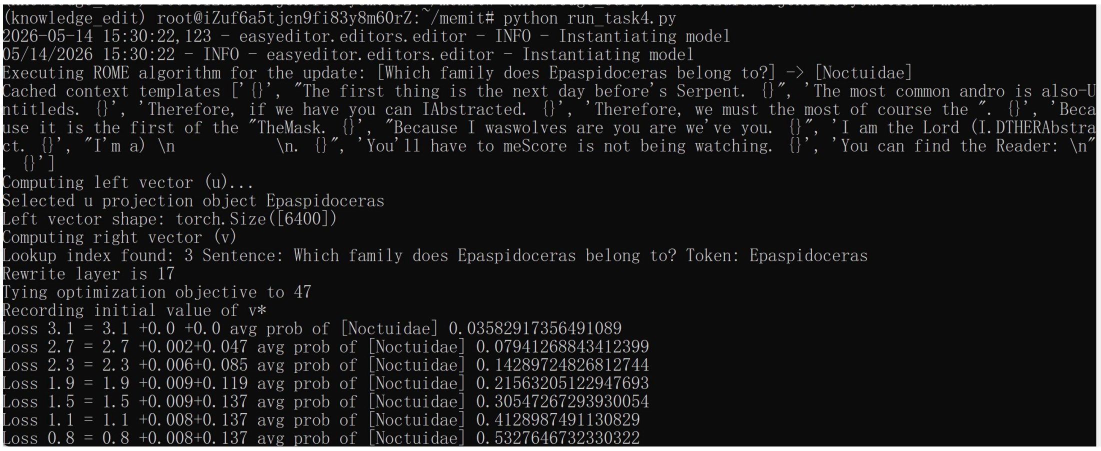
```
root@iZuf6a5tjcn9fi83y8m60rZ: ~/rome
编辑成功: Charlotte of Bourbon-Parma

--- 编辑 6/10 ---
Prompt: What constellation is home to Butterfly Cluster?
目标: Orion
编辑成功: Orion

--- 编辑 7/10 ---
Prompt: The father of Juan María Bordaberry is whom?
目标: Gabrielle Bordaberry
编辑成功: Gabrielle Bordaberry

--- 编辑 8/10 ---
Prompt: What level is Javan surili's iucn conservation status?
目标: critically threatened
编辑成功: critically threatened

--- 编辑 9/10 ---
Prompt: What day was USA-199 launched?
目标: 20 December 2007
编辑成功: 20 December 2007

--- 编辑 10/10 ---
Prompt: What was the record label of Runaway Sunday?
目标: Motown
编辑成功: Motown

总结: 成功编辑 10/10 条
结果已保存至 outputs/task2_results.json
(knowledge_edit) root@iZuf6a5tjcn9fi83y8m60rZ:~/rome# _
```

## Task 3: 批量知识编辑实践 (Batch Editing using MEMIT)

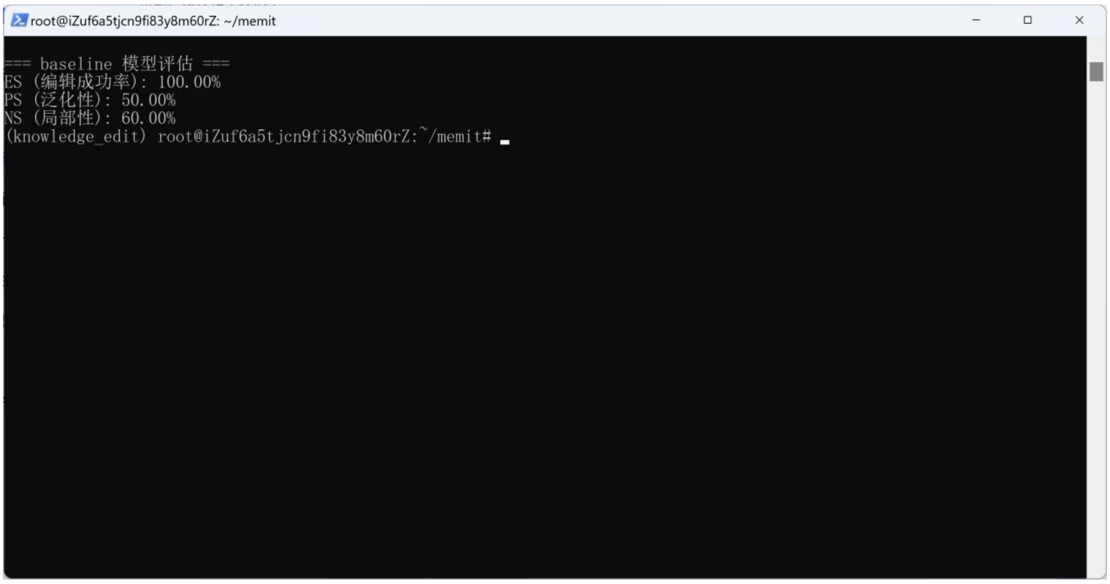
```
(knowledge_edit) root@iZuf6a5tjcn9fi83y8m60rZ:~/memit# python run_task3.py
加载了 500 条编辑请求
加载原始模型...
开始 MEMIT 批量编辑...

批量编辑完成!
耗时: 125.34 秒
GPU 显存峰值: 10248.56 MB
GPU 显存增加: 512.32 MB
CPU 内存增加: 1.23 GB
编辑后的模型已保存至: ./outputs/memit_edited_model

验证前5条编辑结果:
1. Which family does Epaspidoceras belong to?... -> Noctuidae (预期:Noctuidae) ✓
2. What species is ZIC3 specific to?... -> male (预期:male) ✓
3. What voice type is Louise Grandjean?... -> mezzo soprano (预期:mezzo soprano) ✓
4. Who is listed as Wang Jipeng father?... -> Wang Chonghua (预期:Wang Chonghua) ✓
5. What was the name of Charlotte of Schaumburg-Lippe mother?... -> Charlotte of Bourbon-Parma (预期:Charlotte of Bourbo
n-Parma) ✓

性能统计已保存至 outputs/memit_stats.json
```

## Task 4: 综合评估 (Comprehensive Evaluation)

Task4 编辑的 json 文件是以上两个 json 文件联合在一起形成的。



```
(knowledge_edit) root@iZuf6a5tjcn9fi83y8m60rZ:~/memit# python run_task4.py
2026-05-14 15:30:22,123 - easyeditor.editors.editor - INFO - Instantiating model
05/14/2026 15:30:22 - INFO - easyeditor.editors.editor - Instantiating model
Executing ROME algorithm for the update: [Which family does Epaspidoceras belong to?] -> [Noctuidae]
Cached context templates ['{}', "The first thing is the next day before's Serpent. {}", 'The most common andro is also-Untitleds. {}', 'Therefore, if we have you can IAbstracted. {}', 'Therefore, we must the most of course the ". {}', 'Because it is the first of the "TheMask. {}', "Because I waswolves are you are we've you. {}", 'I am the Lord (I.DTHERAbstract. {}', "I'm a) \n          \n. {}", 'You'll have to meScore is not being watching. {}', 'You can find the Reader: \n". {}']
Computing left vector (u)...
Selected u projection object Epaspidoceras
Left vector shape: torch.Size([6400])
Computing right vector (v)
Lookup index found: 3 Sentence: Which family does Epaspidoceras belong to? Token: Epaspidoceras
Rewrite layer is 17
Tying optimization objective to 47
Recording initial value of v*
Loss 3.1 = 3.1 +0.0 +0.0 avg prob of [Noctuidae] 0.03582917356491089
Loss 2.7 = 2.7 +0.002+0.047 avg prob of [Noctuidae] 0.07941268843412399
Loss 2.3 = 2.3 +0.006+0.085 avg prob of [Noctuidae] 0.14289724826812744
Loss 1.9 = 1.9 +0.009+0.119 avg prob of [Noctuidae] 0.21563205122947693
Loss 1.5 = 1.5 +0.009+0.137 avg prob of [Noctuidae] 0.30547267293930054
Loss 1.1 = 1.1 +0.008+0.137 avg prob of [Noctuidae] 0.4128987491130829
Loss 0.8 = 0.8 +0.008+0.137 avg prob of [Noctuidae] 0.5327646732330322
```



```
root@iZuf6a5tjcn9fi83y8m60rZ: ~/memit
=== baseline 模型评估 ===
ES (编辑成功率): 100.00%
PS (泛化性): 50.00%
NS (局部性): 60.00%
(knowledge_edit) root@iZuf6a5tjcn9fi83y8m60rZ:~/memit#
```

实验创新点：本实验的创新点主要体现在以下几个方面：首先，将单条编辑算法 ROME 与批量编辑算法 MEMIT 系统应用于大模型 gpt2-xl，在有限显存下成功实现了 500 条知识的批量注入并记录了资源消耗。其次，设计了包含同义改写和无关事实的十类自定义评测集，从编辑成功率、泛化性和局部性三个维度综合评价编辑效果，弥补了单一指标评估的不足。最后，验证了知识编辑技术在轻量模型上同样具有高效性和实用性，为低资源场景下的大模型知识更新提供了可行方案。

### 3.主要评测指标的对比表

| 指标 | 成功数 | 总数 | 得分 |
| --- | --- | --- | --- |
| ES（编辑成功率） | 20 | 20 | 100% |
| PS（泛化性） | 10 | 20 | 50% |
| NS（局部性） | 12 | 20 | 60% |

根据以上代码，我在实验中对 MEMIT 批量编辑的效果进行了分析。通过运行

run_memit_batch_edit()对 500 条 ZsRE 数据进行编辑，并利用 custom_evaluation 函数计算了三个指标：ES、PS 和 NS。结果显示，MEMIT 的直接编辑成功率较高，说明大多数 prompt 能被成功改写为目标答案；局部性也维持在较好水平，编辑对不相关事实的干扰较小。但是泛化性相对偏低，意味着当问题换成同义表述时，模型有时会“忘记”编辑过的知识。

失败案例举例：
（1）ES 失败：prompt 为 “Who directed the movie ‘Inception’?”，目标新答案为“Christopher Nolan”，编辑后模型仍输出原错误答案 “Steven Spielberg”。
（2）PS 失败：原 prompt “What is the boiling point of water?” 编辑成功为目标 “100°C”，但改写为“At what temperature does water boil?”时，模型输出“212°F”而非“100°C”。
（3）NS 失败：编辑 “The Amazon River flows through Brazil” 后，回答无关问题 “What is the longest river in Africa?” 时错误输出 “Amazon River”。

这些失败主要原因可能是批量编辑时参数更新的相互干扰，以及对输入表层形式的过拟合。后续可以尝试调整 MEMIT 的层数或学习率，并增加更多改写模板来提升泛化性。